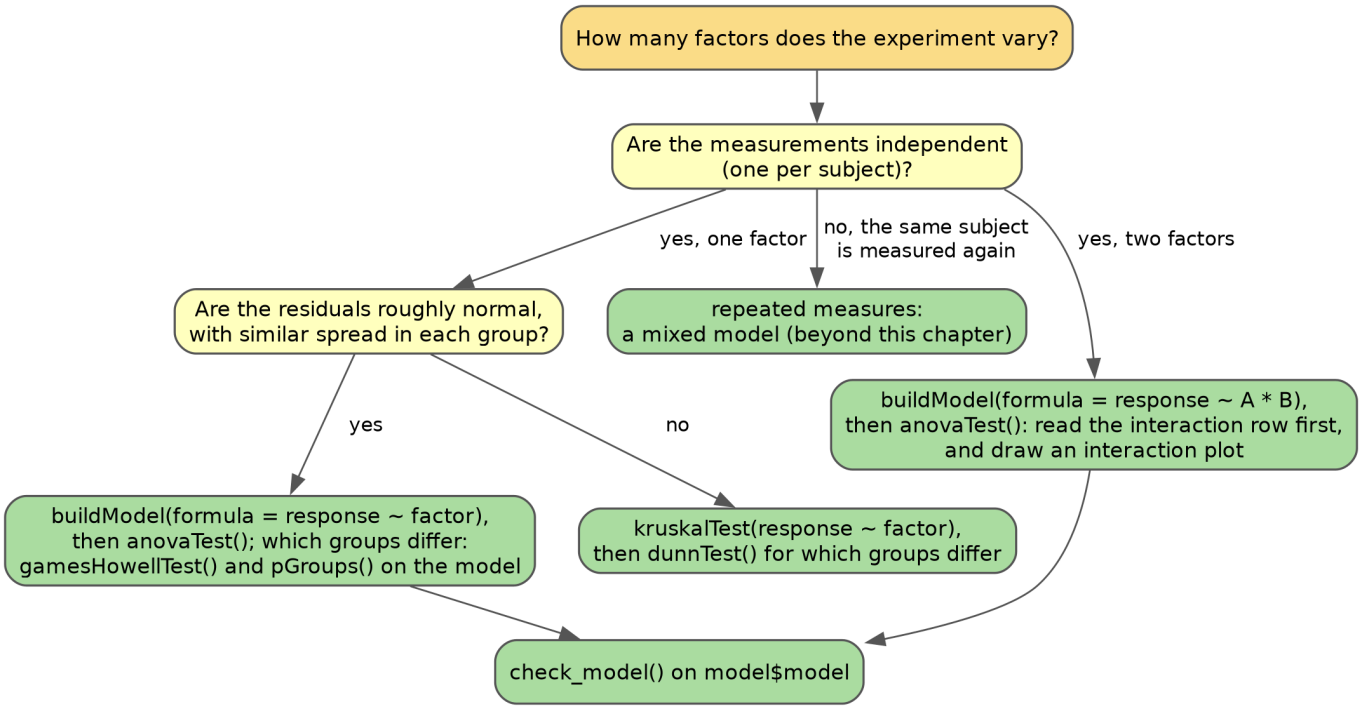
analysis of variance: summary

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An ANOVA splits the variation in a response into the part each term explains and the part left over. **Build the model, then ask it questions:** fit it once with a formula (response on the left, factors on the right, `*` for an interaction), read the ANOVA table, then, for a single factor, use the Games-Howell test on the same model to find which groups differ. **Read the table with the degrees-of-freedom budget in mind:** each measurement adds one unit, a factor with *k* groups costs *k* − 1, an interaction costs the product of its terms' costs, and the rest goes to the residuals. F is each term's variation per unit spent over the residual variation per unit, so it is evidence that a term matters; `ges` is how much it matters. **With two factors, read the interaction row first:** if it is real, a main effect alone can mislead, and an interaction plot shows how. **Check the model** before trusting it: residuals roughly normal with similar spread, and measurements independent (repeated measurements of one subject need a mixed model). For a single factor with non-normal data, use the Kruskal test, then the Dunn test.

Function	What it is for	How it is used in this chapter
`buildModel(formula = )`	fitting the model the ANOVA is read from	`response ~ factor`, or `response ~ factor_1 * factor_2` for an interaction, with `model_type = "linear_regression"`
`anovaTest()`	the ANOVA table: one row per term, with `DFn`, `DFd`, `F`, `p` and `ges`	piped after the model
`gamesHowellTest()`, `pGroups()`	which groups differ, as letters, using each group's own spread	piped after the same one-factor model
`kruskalTest(abundance ~ aquifer_code)`	rank-based test of any difference, one factor	on the data, followed by `dunnTest()`
`check_model()`	diagnostic panels for the residuals	given `model$model`